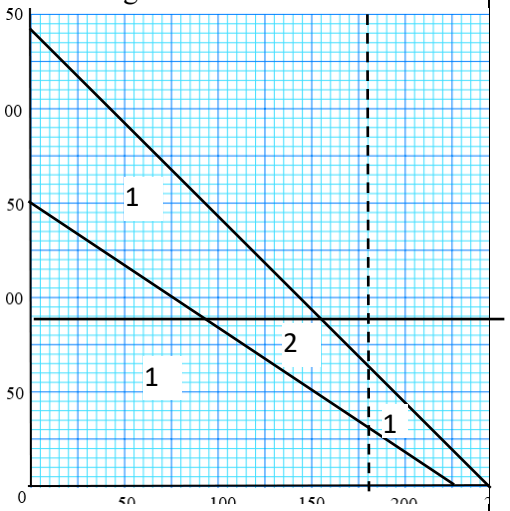
Question	Answer	Marks	Partial Marks	Guidance
7(a)	180 and 240	**2**	**B1** for 180 or for 240	
7(b)	$12x + 18y \geq 2700$ and completion to $2x + 3y \geq 450$	**1**	with no errors seen	After $12x + 18y \geq 2700$ candidate must write $2x + 3y \geq 450$. If any intermediate steps shown they must be correct.

7(c)	$x = 180$ broken straight line and $y = 90$ solid ruled line and $x + y = 240$ solid ruled line and $2x + 3y = 450$ solid ruled line	**B5**	**B1** for $x = 180$ broken straight line **B1** for $y = 90$ solid ruled line **B1** for $x + y = 240$ solid ruled line **B2** for $2x + 3y = 450$ solid ruled line or **B1** for line with a negative gradient passing through (0, 150) or (225, 0)	Penalise first occurrence only of error in solid/broken lines. Condone good freehand lines $x = 180$ and $y = 90$ must touch correct grid line $x + y = 240$ accuracy < 2mm at intercepts extend if necessary $2x + 3y = 450$ accuracy < 2mm at intercepts extend if necessary for B2 and B1 Lines must be long enough to form region
	Correct region indicated  	**B2**	**B1** for region satisfying 3 of the inequalities	If R is not labelled then take the unshaded region as their R. If R is labelled take the lines surrounding R to be the boundaries of their region and ignore shading. Condone solid/broken lines if lines otherwise correct Regions labelled 1 are examples of regions that satisfy 3 of the inequalities but it is not an exhaustive list eg if a line is wrong or omitted. This B1 requires 3 correct (condoning broken/solid) lines to be drawn. (Allow max of one incorrect line to be drawn or one line to be omitted.)

Question	Answer	Marks	Partial Marks	Guidance
7(d)	4200	**2**	**B1** for 150 and 90 or **M1** for *their* 150 × 10 + *their* 90 × 30	Independent of their graph where *their* (150, 90) lies in *their* region For M1, condone use of points on boundary lines eg (180, 60) If R is not labelled then take the unshaded region as *their* R dep on some shading For M1 must either show working or write the co-ordinate/point used for the answer – then we check. For M1 if more than one trial then all points used must be in the region unless there is a selection.